

DEVENDRA CHAUDHARI

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years building and shipping generative AI, agentic AI, and RAG systems to production. Hands-on expertise in LangChain, LangGraph, MCP, A2A, FastAPI, Docker, Kubernetes, and Google Cloud Run. Published researcher with a track record of delivering autonomous multi-agent platforms from prototype to cloud deployment.

EDUCATION

Flinders University

Master of Data Science | Relevant coursework: Statistical Learning, Natural Language Processing, Cloud Computing Adelaide, Australia

Expected Dec 2027

Savitribai Phule Pune University

Bachelor of Engineering in Computer Science | CGPA: 9.13/10.00

Feb 2021 – Jun 2024

Pune, India

WORK EXPERIENCE

AI Engineer

Jul 2024 – Present

Freelance | Remote (concurrent with full-time role Jul 2024 – Present)

- Developed and deployed Generative AI applications including RAG chatbots, autonomous multi-agent systems, and AI-powered automation solutions using LangChain, LangGraph, FastAPI, and Python, reducing average client workflow time by ~35%.
- Engineered intelligent agent workflows with Model Context Protocol (MCP) and Agent-to-Agent (A2A) communication, cutting cross-agent integration time by ~50% compared to custom REST-based approaches.
- Built scalable AI platforms integrating vector databases, embeddings, semantic search, and web tools to support real-time information retrieval and task execution across 4+ client deployments.
- Designed production-ready backend services and REST APIs for AI applications, achieving <200ms median response latency across cloud environments.
- Deployed and monitored AI solutions using Docker, Microsoft Azure, GitHub, and LangSmith, reducing debugging and observability overhead by ~40% through structured tracing pipelines.

Associate Engineer

Dec 2024 – Mar 2025

L & T Technology Services | India

- Engineered enterprise-grade cloud applications leveraging Microsoft Azure services including Azure Virtual Machines, Azure App Services, Azure SQL Database, and Azure AI services for scalable production deployment.
- Architected a scalable Data Access Layer using Entity Framework Core with Database-First design, reducing backend data retrieval time by ~30% and improving system maintainability across 3+ modules.
- Optimized SQL Server performance by designing stored procedures, complex queries, and indexed database schemas, cutting average query execution time by ~45% on high-volume transactional data.
- Automated build and deployment processes by implementing CI/CD pipelines in Azure DevOps, reducing manual deployment effort by ~4 hours per release cycle and enabling continuous integration across development teams.
- Completed advanced Microsoft Azure training encompassing Generative AI, NLP, Machine Learning, Computer Vision, and cloud-native AI application development.

Machine Learning Intern

Feb 2023 – Mar 2023

E-Cell IIT Hyderabad | India

- Built and evaluated machine learning models using Python, TensorFlow, and Scikit-learn to predict seasonal flu vaccine uptake, achieving a ROC-AUC of 0.84 on real-world healthcare datasets.
- Implemented and benchmarked Logistic Regression, Random Forest, and ensemble learning algorithms, selecting the best-performing model through systematic cross-validation and hyperparameter tuning.
- Applied end-to-end data preprocessing including feature selection, normalization, and class imbalance handling, improving model generalization F1 score by ~12% on skewed healthcare data.
- Evaluated model performance using confusion matrix analysis, ROC-AUC curves, precision-recall metrics, and F1 scores, ensuring rigorous and reproducible model assessment.

PROJECTS

AustraliaJobSeeker AI – End-to-End AI Agent Job Platform

May 2026 – Jun 2026

Python, LangGraph, LangSmith, Groq LLM, FastAPI, MCP, A2A, Docker, Kubernetes, Google Cloud Run, CI/CD

- [Live App](#) | [GitHub Repository](#)
- **Real-World Problem Solved:** Seek.com has no visa filter for international students – hours are wasted applying to citizen-only jobs. Built a full production AI agent system that scrapes live Seek jobs and filters them by visa type (500, 485, 482, PR) in real time.
- Architected a LangGraph multi-agent pipeline covering real-time job scraping, job enrichment, visa eligibility checking, resume match ranking, and AI-powered resume and cover letter generation in a single automated workflow.

- Built custom MCP (Model Context Protocol) servers exposing job search, visa check, resume tailoring, and cover letter generation as callable tools for any AI agent, and A2A (Agent-to-Agent) servers enabling true multi-agent collaboration and skill discovery across the system.
- Integrated LangSmith for real-time agent observability and tracing, providing full visibility into every agent decision, LLM call, and pipeline execution in production.
- Deployed on Google Cloud Run with Docker containerization, Kubernetes manifests for production-grade orchestration, and a CI/CD pipeline via Google Cloud Build for fully automated builds and deployments on every git push.

ProjectPilot-AI – Autonomous Multi-Agent System

Mar 2026

Python, LangGraph, LangChain, FastAPI, Streamlit, Ollama, REST APIs

- [Live App](#) | [GitHub Repository](#)
- Architected a production-grade autonomous multi-agent AI platform using LangGraph, enabling dynamic task planning, multi-step LLM reasoning, and real-world automation across 5+ integrated tool agents.
- Engineered a modular AI agent orchestration framework with dedicated planner, executor, and tool-selection agents, streamlining LLM decision-making pipelines and improving end-to-end task orchestration efficiency by ~40%.
- Integrated 6+ external APIs and services – including GitHub API, REST APIs, file management systems, and web browsing tools – enabling fully automated workflow execution without human intervention.
- Deployed cost-efficient local LLM inference using Ollama, eliminating cloud API costs for agent reasoning while maintaining response quality comparable to hosted LLM providers.
- Built scalable RESTful backend services with FastAPI and an interactive real-time frontend with Streamlit, supporting concurrent agent execution and live task monitoring.

YouTube RAG Chatbot

Jan 2026

Python, LangChain, Gemini AI, FAISS, Streamlit, Docker, Kubernetes

- [Live App](#) | [GitHub Repository](#)
- Built and deployed an end-to-end RAG chatbot enabling users to query YouTube video content in natural language using LangChain and Google Gemini LLM.
- Designed a 4-stage RAG pipeline covering video metadata extraction, transcript loading, FAISS vector indexing, and context-aware answer generation, achieving sub-second retrieval latency.
- Generated high-dimensional transcript embeddings using Google Gemini and indexed them in FAISS, enabling semantic vector search across 100K+ token transcripts with <500ms query time.
- Dockerized the full application and authored Kubernetes deployment and service manifests, ensuring cloud-native scalability and production deployment readiness.

RESEARCH PUBLICATION

Book Chapter: “A Real-Time Vehicle Counting System for Detecting Traffic Surveillance” – *Advancing Engineering and Technology through Interdisciplinary Approaches: A Pragmatic Approach*, Cambridge Scholars Publishing, 2026. Presented real-time deep learning-based vehicle detection, classification, and congestion analysis using YOLOv3, OpenCV, and Python for scalable smart traffic management.

Conference: ICATIA-24 – International Conference on Advanced Technologies and Intelligent Automation, May 2024. Presented research on computer vision and intelligent automation systems to an interdisciplinary engineering audience.

TECHNICAL SKILLS

Programming Languages: Python, SQL

AI/ML Technologies: Machine Learning, Deep Learning, Generative AI, Agentic AI, NLP, Computer Vision, LLMs, RAG Pipelines, Prompt Engineering, Fine-Tuning, Embeddings, Vector Search, AI Agent Orchestration, MLOps, Transfer Learning

Frameworks & Libraries: LangChain, LangGraph, LangSmith, TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, FastAPI, Flask, Pandas, NumPy, OpenCV

Databases: SQL Server, MySQL, FAISS, Pinecone, ChromaDB, Vector Databases

Cloud & DevOps: Microsoft Azure, Azure AI Foundry, Azure DevOps, Google Cloud Run, Docker, Kubernetes, CI/CD Pipelines

Developer Tools: GitHub, GitLab, VS Code, PyCharm, Jupyter Notebook

Concepts: MCP Servers, A2A Agents, REST APIs, Microservices Architecture, Model Deployment, Feature Engineering, Hyperparameter Tuning, Model Evaluation

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate (AI-102) – Microsoft – Aug 2025

Microsoft Certified: Azure AI Fundamentals (AI-900) – Microsoft – Aug 2025

Machine Learning with Python – E-Cell IIT Hyderabad – Mar 2023

Problem Solving and Ideation Workshop – Entrepreneurship Development Startup & Innovation Cell – Nov 2021